The hidden margins of participation

A Ghana labor-market paper, a Chile family-economics working paper, and an infrastructure classic from India · two peer-reviewed economics-journal articles plus one World Bank working paper. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

The common thread is that institutions often change the margin on which people participate rather than producing a simple level effect. In Ghana, psychological distress predicts reluctance to accept work outside the home but not productivity once an at-home job is offered. In Chile, divorce legalization changes the insurance value of marriage through asset division and household specialization. In India, roads move workers out of agriculture without quickly generating broad village-level income or production gains. **Literature trend:** the strongest applied work increasingly asks which behavioral margin a policy unlocks—entry, risk sharing, or reallocation—and then designs evidence around that margin. This is more informative than treating work, marriage, or infrastructure as single outcomes with uniformly positive or negative effects.

01 CURRENT / MENTAL HEALTH AND LABOR-MARKET PARTICIPATION · STARTS ON PAGE 2 · Journal of Development Economics 181 · 2026 · peer reviewed · World Bank open edition

Psychological Barriers to Participation in the Labor Market: Evidence from Rural Ghana

Does poor mental health reduce the ability to work, or chiefly the willingness to enter socially demanding jobs? In rural Ghana, a two-phase design first compares take-up of identical attractive work offers at home and at an outside worksite, then randomizes at-home offers among willing workers. Depression and anxiety strongly predict lower take-up outside the home but not at home; once an at-home offer is randomized, mental health does not predict completion, earnings, productivity, or quitting. The phase-one association is not a fully randomized mental-health effect, and the lean-season setting, gender mix, and sample limit generalization. The result matters because remote work may bypass participation costs without requiring a productivity deficit to be repaired.

02 CURRENT / FAMILY ECONOMICS AND LEGAL INSTITUTIONS · STARTS ON PAGE 17 · World Bank Policy Research Working Paper 11475 · September 2026 · R&R; at JDE

Untying the Knot: Divorce Legalization and the Decline of Marriage

Did Chile's 2004 divorce reform weaken marriage, or make it a better insurance contract? Population trends and a dynamic difference-in-differences design compare women married under community property, where equal division changed the stakes, with women under separate property. Marriage, marital fertility, and community-property use recover after legalization; female labor supply falls among young, highly educated women in community-property marriages. A limited-commitment model links the response to household assets and risk sharing. Nationwide trend breaks are less causal than the regime comparison, and property-regime choice and parallel trends remain important assumptions. The paper reframes exit rights as potentially raising the insurance value of marriage rather than simply making marriage less durable.

03 CLASSIC / INFRASTRUCTURE AND STRUCTURAL TRANSFORMATION · STARTS ON PAGE 32 · American Economic Review 110(3) · 2020 · peer reviewed classic · World Bank open edition

Rural Roads and Local Economic Development

Do paved rural roads transform village economies? India's PMGSY program assigned road eligibility at population thresholds, creating a fuzzy regression discontinuity that the authors link to household and firm censuses and remote-sensing data. Roads reduce agricultural employment by about 10 percentage points and increase manual labor, but show little short-run improvement in consumption, assets, or agricultural production; local firms expand only modestly, suggesting much of the new work lies outside the village. Effects are local to threshold-complier villages and a medium-run horizon, with possible spillovers and measurement limits. This classic is a useful corrective: connectivity can enable labor reallocation without automatically creating broad local prosperity.

Editorial note: references and appendices were omitted where possible; no included article contributes more than about 15 text pages, excluding table-heavy pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

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Policy Research Working Paper 11271

Psychological Barriers to Participation in the Labor Market

Evidence from Rural Ghana

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Policy Research Working Paper 11271

Abstract

Mental health conditions are strongly associated with reduced labor market participation, but the underlying channels through which such conditions impact labor supply remain unclear. This paper reports on a two-phase study decomposing this relationship by examining (i) job take-up decisions; (ii) labor supply, output, and earning conditional on job take-up; and (iii) quit rates. In Phase 1, women in rural Ghana were asked whether they would be willing to take up a cash-for-work job during the lean season when alternative work is scarce. The findings show that individuals with depression and anxiety, which are common in this population, are much more likely to decline work offers outside the home but equally likely to accept work-from-home positions. In Phase 2, jobs at home were randomly offered to those who were willing to work from home, avoiding selection effects. Neither depression nor anxiety predicted work completion, income, or quit rates. These findings suggest that poor mental health may harm labor market outcomes in traditional jobs outside the home via reduced take-up, above and beyond the established negative impacts of mental health on productivity in work outside the home. The results also suggest an alternative approach to improving labor market outcomes for those in poor mental health: work-from-home opportunities, which are not associated with lower take-up or lower productivity on the job for those in poor mental health.

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Produced by the Research Support Team

1 Introduction

Poor mental health is prevalent in low-income populations around the world and has been associated with low labor supply and earnings, creating the potential for negative feedback loops which may perpetuate poverty (Barker et al., 2022; Biasi et al., 2021; de Quidt and Haushofer, 2016; Fuhr et al., 2019; Hakulinen et al., 2019; Lund et al., 2024; Mojtabai et al., 2015; Patel et al., 2011; Ridley et al., 2020; Weobong et al., 2017; World Health Organization, 2022b). While treating mental illness can have significant positive impacts on labor supply, these effects are not universal, with more muted or null results among some populations (Angelucci and Bennett, 2024; Baranov et al., 2020; Barker et al., 2022; Bhat et al., 2022; Lund et al., 2024; Patel and Kleinman, 2003; Patel et al., 2017).

These mixed results may have many potential drivers and relatively little is known about the channels through which poor mental health reduces labor supply and income. In order to better understand how mental health conditions affect these important outcomes and begin to develop policy solutions, we decompose the process and study three of its key components: (i) job take-up; (ii) labor supply, output, and earnings conditional on job take-up; and (iii) quit rates.

To accomplish this goal, we begin by measuring mental health in a low-income population of women in rural Ghana. We focus on women in a low-income setting as they suffer disproportionately from mental health conditions and have lower and more variable labor force participation (World Health Organization, 2022b). Consistent with global patterns, mental health in this population is poor, with 39.7% of individuals suffering from depression and 45.0% of individuals experiencing anxiety, relying on the "moderate" cutoffs for the corresponding mental health scales —the Patient Health Questionnaire (PHQ) and the General Anxiety Disorder scale (GAD) (Kroenke et al., 2001, 2003; Spitzer et al., 2006).

Next, to avoid selection due to search or endogenous job offers and isolate later portions of the causal chain, in Phase 1 of the study we make attractive and identical job offers to these individuals. The work offered is similar to cash-for-work opportunities often provided by governments with the aim of supporting the poor during the lean season in that it is part-time, low-skill, and offers a wage premium of roughly 50%. Further, the decision to work is high stakes, with the income provided by the job accounting for about

46% of average total household income during this period. We elicit willingness to work under two different scenarios: work from home or work from a common worksite.[1] This variation speaks to a key feature of depression and anxiety: those in poor mental health frequently seek to avoid new or potentially stressful situations. Allowing individuals to take-up work from a familiar environment may alleviate key stressors associated with labor force participation and increase take-up.

Our first key finding is that poor mental health (both anxiety and depression) is strongly associated with a reduced willingness to take up this attractive work opportunity. As can be seen in Figure 2, participants who are anxious and depressed (i.e. scoring above the PHQ and GAD thresholds) are roughly 54% more likely to decline the offered work than those who are neither anxious nor depressed (11.9% vs. 7.7%). These differences are not due to existing work obligations, with more depressed individuals being less likely to cite existing work as a reason to decline and more likely to cite inability or unwillingness to take up the work as the reason for refusal. Nor do these differences in take-up appear to be driven by childcare responsibilities: respondents in poor and good mental health have similar numbers of young children, and results control for household demographics.

However, this strong association between depression and anxiety and the willingness to work is only present in work offered outside the home; correlations between either aspect of mental health and willingness to work from home are quite small in magnitude and statistically insignificant (Table 5). This stark difference in take-up rates across work location for those in poor mental health is consistent with many of the symptoms of depression and anxiety —avoidance behaviors, low energy levels, or nervousness— which could make individuals less willing to leave the home despite the relatively generous wages.

These results suggest mental health is associated with reduced job take-up and low earnings in many contexts where work is primarily found outside of the home. This reduced willingness to take up work outside the home when in poor mental health may exacerbate known impacts of poor mental health on productivity at work in traditional (outside the home) jobs (Lund et al., 2024; Ridley et al., 2020). This existing body of evidence raises the question of whether poor mental health may also impact labor outcomes in work from home by impacting performance on the job, even though individuals

[1] The location of the worksite was either in the participant's village or a nearby village and participants were responsible for their own transportation, which is often by foot.

in poor mental health are equally likely to take up these jobs.

To answer these questions, we draw on Phase 2 of the study in which we enroll a random sample of women who are willing to work from home and randomly divide this sample into two groups: those who are offered to work from home (treated) and those who are not offered any work (control).[2] The randomization also allows us to examine whether, conditional on being willing to take up a job, this random job offer is equally valuable to depressed and anxious individuals. In other words, among those willing to take up the work, do those with poor mental health have similar labor supply, output, earnings, and quit rates?

To study these channels, we examine the labor supply, output, and earnings of participants who were randomly provided with jobs, including heterogeneity by baseline mental health which is, on average, quite poor in this population (34.9% and 32.8% of the sample experience at least moderate depression and anxiety at the baseline of Phase 2, respectively, based on standard cutoffs in the literature). Participants report bi-weekly on days worked and earnings in surveys. These data are combined with administrative data on work completed as a part of the study.

Unsurprisingly given that little work is available in the lean season, this random job offer greatly increases the likelihood of work among those offered the job, nearly quadrupling engagement with self-reported wage labor from 17% to 60% over the three months of work offered. We find that, although we cannot rule out small effects, in this group depressed and/or anxious individuals are just as likely to supply labor and produce as much as those in better mental health, and correspondingly earn similar amounts (Table 7). Further, poor mental health is associated with a small and marginally significant delay (improvement) in the time to quitting but does not predict quit rates overall.

In short, we conclude that depression and anxiety impede labor market success not only via lower productivity in traditional work contexts, but also because those with mental health conditions are much less likely to take up new labor opportunities. However, conditional on taking up work that can be done from the home —which is uncorrelated with baseline mental health— mental health does not impede labor supply, output, or

[2]We did not offer jobs outside the home both for logistical reasons and because those taking up the jobs are likely to be highly selected given the prior results. Further, as individuals cannot and should not be compelled to work, examining outcomes conditional on willingness to work is the more policy-relevant and ethical approach.

even when attractive job opportunities are directly offered, poor mental health —which is highly prevalent in this context— substantially reduces take-up in traditional employment opportunities outside the home, suggesting an important but previously overlooked constraint on engagement with the labor force. Further, although only suggestive, our analysis of respondents' decisions regarding household members' willingness to work highlights the potential for the effects of poor mental health to extend beyond the impacted individual.

Finally, we inform the literature on cash-for-work programs, which are widely used as anti-poverty tools in low-income countries (Bertrand et al., 2021; Besley and Coate, 1992; Imbert and Papp, 2015; Murgai et al., 2016). These programs typically operate through standing job offers outside the home, assuming that those most in need will self-select into the work. Our findings suggest that poor mental health —which is disproportionately prevalent among the poor— may limit engagement with these programs precisely among those they are intended to help. This has important implications for program design: screening for mental health conditions and providing complementary mental health interventions, or innovating to provide other targeted intervention designs such as work from home arrangements, may be necessary to achieve the full potential of cash-for-work as an anti-poverty tool.

The remainder of the paper begins with background on the context and study design in Section 2. We outline our conceptual framework in Section 3. The empirical approach is described in Section 4. Section 5 provides results, and Section 6 concludes.

2 Context and study design

Our study is conducted in northern Ghana, in rural areas outside the city of Tamale. The population is heavily agricultural and the poverty rate in this area is high, with over 70 percent of people defined as multi-dimensionally poor in the study districts (Ghana Statistical Service, 2023). Similar to many other areas with high poverty rates, mental health conditions are relatively high in the region: Ae-Ngibise et al. (2023) observe a prevalence of probable depression of 15.6%, and Barker et al. (2022) find that about 58% of the general population suffers from symptoms associated with some degree of psychological distress (compared to about 13 percent in the United States according to Dhingra

et al., 2011).[3]

Timing – The study was conducted in the lean season, after planting and before harvest of the major staple grain, maize. The lean season offers limited work opportunities, especially opportunities for wage labor. Because many households are running low on grain and have limited income, this time is typically a period of relatively high stress. The study occurred in two phases over a period of four months. Phase 1, which occurred in the first month, examined the participants' willingness to take up new work opportunities both at worksites and from home. Phase 2, which spanned three months following Phase 1, offered randomly selected individuals from the group willing to work from home the opportunity to work from home stitching bags as described below. Phase 2 was divided into six 2-week periods with data collection on mental health, labor supply and earnings collected in face-to-face interviews at baseline and endline (following period 6), and via phone surveys at the end of each period in-between, resulting in 5 total phone surveys.

Work opportunity – We offer to randomly selected participants in Phase 2 the opportunity to stitch bags —allowing us to observe output each period— at their home. Prior to beginning their work, participants are trained over 3 days by professionals to ensure they are able to meet the minimum quality standards. Materials for the bags are dropped off at the beginning of each two-week period and the finished bags are picked up three times throughout the period. Payment is piece-rate (GHS 12 per completed bag) and made immediately upon the pickup of the finished product. The time required depends on the amount of work offered each period, but on average it is about 3 to 4 days of work per period and ranges from 1.5 to 6 days per two-week period.

The jobs offered are an important potential source of income in this context. Because work opportunities are often limited during the lean season, the average household income at the beginning of Phase 2 is GHS 474 (USD 32.7) per month, equivalent to GHS 237 per 2-weeks period. The majority of this income typically comes from male adults, who earn more per hour and work outside the home more frequently. Participants are offered

[3] Ae-Ngibise et al. (2023) define "probable depression" as the combination of a PHQ-9 score greater than four and reporting at least two cardinal symptoms of depression or anhedonia (loss of interest or pleasure). Barker et al. (2022) construct an index based on three measures: the Kessler Psychological Distress Scale, a self-rating of mental health taken from the Behavioral Risk Factor Surveillance System (BRFSS) and a self-report of days in the month without poor mental health.

an average of GHS 108 (around 7.5 USD) each two-week period. The job offers have the potential to increase total household income by 45.6% on average given the low labor demand outside the study. However, it is also important to note that, as is typical during lean seasons, consumption is significantly higher than income during this time such that the job offer is roughly equivalent to 14% of household consumption. The hourly wage premium is designed to be similar to many cash-for-work programs such as India's Mahatma Gandhi National Rural Employment Guarantee Scheme (NREGA), and is roughly 50% higher than the prevailing day labor rate for women (Nagaraj et al., 2016).[4]

The fact that we offer the same jobs uniformly to all participants allows us to address some of the potential areas of concern around confounding. Specifically, this approach allows us to: 1) abstract away from differences in search behavior which could generate selection into who is observed, and 2) hold constant the features of the job which might otherwise endogenously (and unobservably) vary with the individual's characteristics such as their mental health status.

Phase 1 – This first phase starts with the recruitment of participants in 30 rural communities with limited work opportunities during the lean season. The communities are distinct villages with 115 households on average. In each community, we target households using a random walk procedure. To be eligible, households must have access to a phone in order to respond to phone surveys —though nearly all did— and have five or fewer adults and a total household size of less than 16 members.[5] The random sampling in the targeted communities results in 838 eligible households. However, because Phase 2 was limited to women, we also limit Phase 1 analysis to women for consistency. Of the 838 eligible households, 514 have female respondents.[6] Twenty-three of the female respondents are individually ineligible due to age (<18 or >65) and are not asked to

[4]The wage premium depends on the assumed number of hours required per bag, estimated by the stitching trainers, and ranges from 20% to 80% depending on the assumptions used.

[5]The household size was set to ensure that we could survey the complete households in a reasonable time, and because in the second phase, we aimed to offer jobs that would substantially improve the economic situation of the participating households. This criterion screened out 366 households (30.3%) of the 1,202 households originally contacted.

[6]If both the household head and his spouse are present, one of them is randomly selected to be the main respondent. If either the head or his spouse is absent, the enumerator randomly selects one of the present adult household members to respond. This sampling procedure was used as this was a pilot study for a future RCT and we were considering adding men to the full RCT the following year.

provide information about their willingness to work. The remaining 491 eligible respondents are considered in our "willingness-to-work sample", used to assess the relationship between a respondent's mental health and their willingness to take-up new job offers.

In each household the respondent is then asked to provide basic socio-demographic information about the household. They are then read a description of the work opportunity and asked if they would be willing to do such work, both for work from home and work outside the home.[7] We randomize whether the respondent is asked first about work from home or work from a worksite, and control for an order dummy in all specifications despite the fact that there is no significant impact of the order in which the questions were asked. Respondents are aware that not all individuals would be offered the work, but that it would only be offered to those interested in taking it up. Next, respondents are asked to report whether other adults in their household would be interested in the job or not, again responding both for work from home and for work outside the home. Finally, participants also respond to the PHQ-2 screening tool, assessing depression, and the GAD-2 module, assessing anxiety. The screening criteria, data collection, and sample sizes are summarized in Table 1.

Phase 2 – Study participants in this phase are first sampled through the process described for Phase 1 participants. To be eligible, individuals must be non-pregnant women aged 18–65 years who are willing to work from home. In Phase 1, they either indicated their willingness to work directly (N=491), or another individual in the household indicated wether they would be willing to work (N=1076).[8] Among these 1,567 women, 1,447 are willing to work from home. To select participants for Phase 2, we first selected households with at least one eligible woman who was willing to work and then randomly sampled 400 of these households. Among the sampled households, if there was more than one eligible woman willing to work, one woman per household was randomly selected to

[7]The job offer is described as: "We are conducting this part of the survey to identify individuals who might be willing to participate in the study on people's economic circumstances during the lean season. You have been randomly selected to answer this part of the survey about yourself and other household members. If a member from your household is selected to participate in the study, the person will be given a gift for subsequent surveys they answer. In addition, some of the study participants will be offered work for a period of 12 weeks; others will not be offered any work. Study participants who are offered work will work stitching bags. Each bag will be paid 12 GHS. No previous experience is required. We will provide training. Note: if your household is selected, only one member from your household will be selected to participate." This portion of the survey was prior to, but separated from, the mental health measures by other content (e.g. questions regarding gender norms).

[8]Of the individuals for whom another individual responded, 500 were women and 576 were men.

be enrolled. We were unable to recontact 10 of these individuals because they had left their village. Because they were unaware of their assigned arm prior to dropping from the study we omit them. The remaining 390 completed an informed consent and were enrolled, constituting the final study sample for Phase 2.

Following the random selection, we return to the households to conduct a baseline survey with the Phase 2 participants. The baseline survey includes measures of labor supply, income, consumption, and mental health (PHQ-8, GAD-7, Cohen's Perceived Stress Scale, and the Penn State Worry Questionnaire). Additionally, the participants are asked to confirm again their willingness to work, which all did. The 390 individuals completing this baseline survey are considered our "interested sample".[9] Then, of the 390 "interested" participants, we randomly make job offers to work from home to 320 of these participants (our treated group) and not to the other 70 participants (our control group).[10]

Both treated and control participants in Phase 2 are interviewed by phone every two weeks during the study. Both groups also have a final in-person interview (endline survey) once the work is completed. During the bi-weekly phone interviews we elicit labor supply, income, and consumption measures as well as the participant's mental health (GAD-2 and PHQ-2). Additionally, treated participants have administrative data collected on their completion of bags each period, noting both the number and quality as well as the payment made for the work. The endline interview includes the same measures of labor supply, income and consumption, and the mental health measures used in the baseline. Attrition over the course of Phase 2 was minimal: only six participants did not complete the endline survey (3 in each arm, see Table A.1).

[9] In a given household, the participant from Phase 1 is not necessarily the same person as the "interested" participant from Phase 2. They overlap in 156 cases. This is because the Phase 2 participant was randomly selected among all eligible women from Phase 1 who signaled interest in the job, whether they were the Phase 1 respondent herself or not.

[10] The experiment in phase 2 was a pilot study for a larger experiment to be conducted the following year studying the effects of income volatility and uncertainty. Following the anticipated design for the full study, treated participants were all offered the same type of work (stitching bags from home). However, expected workload (and hence income) and forward-looking knowledge of that workload over time varied across arms: one group had the same income in every period, the second group had variable income across time that was known in advance, and the last group had both variable and uncertain income. These differences were not described in the brief description of the work during Phase 1 and hence could not have impacted take-up decisions. No participants declined to participate in Phase 2 after learning these details during the informed consent. Because the differences in the timing of the workload are not relevant to the questions of interest in this paper and the results are statistically indistinguishable across treated arms, we aggregate these groups together in our analyses resulting in a treatment group that is substantially larger than the control.

Measurement – Mental health is proxied by depression, assessed using the Patient Health Questionnaire (PHQ), and anxiety, assessed using the General Anxiety Disorder Questionnaire (GAD) (Kroenke et al., 2001, 2003). Both the PHQ and the GAD are among the most frequently used scales to screen for depression and anxiety globally, and have been validated and used previously in Ghana (Adjorlolo, 2020; Anum et al., 2019; Barthel et al., 2014; Weobong et al., 2009). We use the 8-item version of the PHQ and the 7-item version of the GAD at the baseline and endline in Phase 2, and the 2-item version of the PHQ and GAD in phone surveys and Phase 1.[11] The local language has no colloquial written form. Hence, to ensure consistent administration of the items and fidelity to the scales in this context we discussed appropriate approaches with a local expert, Dr. Ben Weobong. The final protocol included translating each question in the survey into the local language, back translation (via a recording) from a second individual blind to the original scale, editing for consistency as required, and then generating a final recording of the item. The enumerator explained the scale, played the recording of each item one by one, and recorded the participant's responses.[12]

The main outcomes of interest are the women's willingness to work (Phase 1), and their labor supply, output, earnings, and quit rates (Phase 2). Willingness to work, both for the respondent and other household members, is captured in Phase 1 as explained above. Participants in the "interested sample" in Phase 2 report their labor supply and earnings over the past ten days in all surveys. They are also asked to report the labor supply and earnings of other household members over the age of 12 for the same period. Not all participants include their study earnings in their reported earnings. In the main tables, we correct the reported earnings to always include the study earnings. The study work is organized in periods of two weeks and the survey corresponding to each period is administered in the last four days of the two-week period. The ten-day recall therefore allows us to capture labor and income during that period only and does not cover previous periods.[13]

[11] We follow established clinical thresholds for interpreting these scales. For the PHQ-2 and GAD-2, scores of 3 or higher (out of 6) indicate probable depression and anxiety respectively (Kroenke et al., 2003; Plummer et al., 2016). For the PHQ-8, we use the threshold of 10 or higher to identify potential depression (Kroenke et al., 2009). For the GAD-7, scores of 10 or higher indicate moderate anxiety requiring clinical attention (Spitzer et al., 2006).

[12] Although use of recordings is not standard practice for these scales, the accuracy and reliability of the administration combined with strong comprehension in earlier pilots suggested they were the optimal approach for the local context.

[13] While the 10-day recall omits 4 days of the period, the timing of the survey is random across treated

measures of mental health, and with different sets of control variables. Additional details of these robustness checks are provided in the results section.

4.2 Phase 2 – Mental health and the effects of job offers

In the second phase, we randomize job offers among the participants who are willing to work from home. We estimate the effects of the job offers on labor supply, output, earnings, and quit rates as well as whether mental health mediates the effects of these job offers. Specifically, we estimate:

$$\begin{aligned} Y_{i,t} = \gamma_0 + \gamma_1 \text{Job offer}_i + \gamma_2 \text{Mental health}_{i,0} + \gamma_3 \text{Mental health}_{i,0} * \text{Job offer}_i \\ + T_t + \boldsymbol{X}_i \boldsymbol{\gamma}_4 + \zeta i, t \end{aligned} \tag{2}$$

Where $Y_{i,t}$ is outcome Y for individual i in period t, "Job offer" is equal to one if i was randomly offered a job, "Mental Health" is one of the mental health variables measured in period 0 (the baseline), and T_t are period fixed effects. $\boldsymbol{X}$ controls for the same demographics as in Eq. 1, as well as controls for which individual in the household responded in Phase 1, but omits the controls for order effects and presence of another household member as they are not relevant in Phase 2. When we estimate Equation (2), we cluster standard errors at the individual level given the multiple periods. As with the Phase 1 analysis, we also examine the robustness of these estimates to a number of variants such as binary measures of mental health, use of the short form measures of mental health, and changes to the control set.

5 Results

5.1 Phase 1: Mental health and willingness to work

Both anxiety and depression are strong predictors of the willingness to work pooling across locations, with marked differences in refusal rates among those with greater anxiety, greater depression, or both, compared with individuals in good mental health (Figure 2). Specifically, refusal rates are 1.3 (2.9) percentage points or 17.4 (29.7)% higher among participants with high anxiety (depression). Being identified with both high depression and high anxiety is associated with refusal rates that are 4.16 percentage points or 54.0%

higher.

In Table 5, we test the statistical significance of these associations, this time distinguishing between working outside the home (top panel) or from home (bottom panel). We find that the PHQ-2 and the GAD-2 strongly correlate with the decision to accept a job offer if the work must be done at a worksite, but not if it can be done from home. On average, an increase of one standard deviation on the PHQ-2 scale (1.68 points) is associated with a reduction of 3.0 percentage points in the likelihood of accepting the offer at a worksite. The association with the GAD-2 (SD = 1.64 points) is of a similar magnitude. The association between the index of mental health and willingness to take up work is slightly larger in magnitude, 3.7 percentage points, but not significantly different than the individual measures of mental health.

These results are in strong contrast to the associations between mental health and willingness to take up work from home. The point estimates when considering work from home are both small in magnitude —none larger than 0.4 percentage points— and variable in sign. These correlations suggest that the relationship between mental health and the take-up of employment opportunities may vary substantially with the context in which the work is provided and may exacerbate effects in low-income settings where remote work is less common.

5.1.1 Why are individuals unwilling to work?

To further understand these associations, we asked participants who were not willing to work why they did not want to take up the job. In Table 6, we correlate the mental health measures with the reasons given for refusing the work offer. Each cell presents the coefficient from a distinct regression, pooling across work locations. Notably, people with higher depression and anxiety scores are significantly less likely to say that they already have paid work as a reason for refusing job offers. Much of this difference appears to be accounted for by a positive association between poor mental health and being unable or unwilling to work, though these associations are less precise and the relationship is only statistically significant at the 10% level for depression. The correlations between mental health and other reasons such as a lack of transportation or student status are generally weaker and not statistically significant.

The evidence that the lack of take-up is not driven by better work opportunities is also

We begin by estimating the overall impact of a job offer on each of these outcomes in Column 1, findings large effects on work engagement, number of days of worked, and earnings, as noted earlier.[16] We then turn to our primary question of interest: whether the impact of a job offer varies by mental health status. The mental health status measures are standardized baseline measures of depression (Col. 2), anxiety (Col. 3), and an average of the two standardized measures (Col. 4). The interaction term in equation 2 allows us to test whether the participants' baseline mental health moderates the effect of the job offers.

The interaction terms are relatively moderate in magnitude and are never statistically significant. Take for example the effect on the extensive margin shown in the top panel. The coefficient on the interaction term is -0.036, which represents roughly 8.3% of the main effect of being offered a job. Similarly, for earnings, the coefficient on the interaction term in column 4 is -5.35 GHS compared to a mean increase of 102.72 GHS when offered a job. Turning to bags produced, we find no evidence that poorer baseline mental health is associated with lower productivity, with coefficients that are both positive and negative and never statistically significant. We cannot reject the hypothesis that productivity is similar across levels of mental health. This similarity in outcomes across those in good and poor mental health is also visually evident in Figure A.1.[17] However, the limited sample size also means we cannot rule out modest effects on these outcomes. Results are similar when we use instead binary measures for mental health (Table A.6), short-form measures of mental health (PHQ-2 and GAD-2, Table A.7), or when demographic controls are omitted (Table A.8).

5.2.2 Quit rates

Finally, to examine the relationship between mental health and work retention, Table 8 reports correlations between mental health indicators and quit rates among those offered a job stitching bags. We use two measures of quitting: (i) non-submission of bags in the final period and (ii) a variable indicating the last period in which the participant

[16]Bag completion data in Figure A.1 is administrative but the work data in Table 7 is self-reported. Notably, this results in a gap between reported levels of work in the two datasets. We believe some women did not classify bag-stitching as "work" since it occurred at home and outside their usual activities. However, to be conservative, we retain self-reported work status in Table 7.

[17]We also explore the possibility that effects may vary with the work schedule the participant was assigned to (stable, predictable volatility, or unpredictable volatility) but find no differences.

ternative employment opportunities (Nagaraj et al., 2016). The fact that mental health significantly correlates with job take-up suggests that integrating mental health support services into existing workfare programs could increase their effectiveness. Further, because mental health is associated with job take-up when the job must be done outside, but not when one can work from home, this indicates that more flexible work arrangements can be a promising way of including people with poor mental —who often represent a substantial share of the population in low-income settings. As the output associated with many existing workfare programs is already limited, altering production tasks for home-based work may not have large productivity costs. Further, the ever-increasing reach of internet connectivity may continue to expand opportunities for productive work from home opportunities even for those with limited education. For example, workers could undertake data labeling tasks to train AI or engage in tasks with both personal and community benefit that could be verified via phone photographs (e.g. digging a drainage ditch in front of their home with before and after photos). Moreover, given that poor mental health appears to create barriers to accepting work opportunities even when little other work is available, policymakers may need to reconsider the assumption that those most in need will naturally self-select into these programs.

However, a work-from-home approach has tradeoffs beyond the added complexity. For example, traditional work outside the home might offer additional benefits through social interaction and network building that could improve mental health outcomes. But emphasizing such arrangements could create equity concerns: some individuals might benefit from increased social interaction while others with mental health conditions would be excluded entirely, missing both potential social benefits and economic opportunities. In short, this research suggests that the potential for work-from-home arrangements to benefit those in poor mental health is high, but more research is also needed to fully flesh out both logistical challenges and potential trade-offs.

Finally, our results point to important directions for future research. While we document that mental health is strongly associated with willingness to work outside the home, understanding the specific psychological mechanisms behind this reluctance —whether related to self-efficacy, misperception, or other factors— remains an important area for investigation. Additionally, examining whether similar patterns hold in other contexts, with other types of work, piece rate versus wage work, or lower or higher-paying work

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Untying the Knot

Divorce Legalization and the Decline of Marriage

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Policy Research Working Paper 11475

Abstract

This study investigates the impact of divorce legalization on marriage decisions and intra-household specialization using Chile's 2004 reform as a case study. Opponents feared that introducing divorce would weaken the commitment value of marriage. Instead, the study documents a reversal of sharply declining trends in marriage rates, marital fertility, and the choice of the community property marriage regime in the 15 years following the law. These effects were most pronounced among highly educated women. The study hypothesizes that the introduction of legal divorce with equal division of marital assets allowed husbands to insure homemakers against the economic risks of informal marital separation, thereby incentivizing specialization and fertility. In support of this mechanism, the study shows that the law caused a large reduction in the labor supply of young women married before the law under community property compared to those married under asset separation, who were not affected by equal asset division. Furthermore, the findings show that the concentration of these effects among the highly educated is driven by their higher levels of household assets, as predicted by simulations from a limited-commitment, collective model of marital investment.

This paper is a product of the Development Research Group, Development Economics. It is part of a larger effort by the World Bank to provide open access to its research and make a contribution to development policy discussions around the world. Policy Research Working Papers are also posted on the Web at http://www.worldbank.org/prwp. The authors may be contacted at cjoubert@worldbankgroup.org. A verified reproducibility package for this paper is available at http://reproducibility.worldbank.org, click **here** for direct access.



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Produced by the Research Support Team

> Divorce is probably of approximately the same date as marriage. I believe, however, that marriage is a few weeks older.
>
> *Voltaire*

1 Introduction

Recent empirical work shows that women worldwide incur large lifetime earnings penalties because of childbearing and career interruptions (Kleven et al., 2025). Because sharing the cost of parenting equally between the sexes remains difficult, marital fertility often requires a credible long-term commitment: a husband's ability and willingness to compensate his wife for the foregone labor market opportunities. However, the possibility of divorce is thought to undermine this mechanism and discourage marriage-specific investments, such as childbearing (Rasul, 2006). Indeed, evidence from unilateral-divorce reforms links easier access to divorce to declines in marital investment: lower fertility (Stevenson, 2008; Bellido and Marcén, 2014), reduced marital specialization and shifts in spouses' labor supply (Johnson and Skinner, 1986; Parkman, 1992; Gray, 1998; Genadek et al., 2007; Stevenson, 2008), as well as lower educational investment and homeownership (Stevenson, 2008).

In this study, we examine the impact of divorce legislation on marital decisions in the context of a more radical change: the legalization of divorce, which was not permitted in Chile until 2004. Besides allowing for the dissolution of marriage, the new law provides for an equal division of assets and regulates alimony. In line with family economics theory, opponents of divorce legalization in Chile argued that it would reduce the appeal of marriage by weakening its commitment value (Cox, 2011). More broadly, this argument underlies policy discussions surrounding the liberalization of divorce along various dimensions (e.g., unilateral divorce, no-fault divorce, and mandatory separation periods). This study documents patterns indicating a large effect in the opposite direction, rationalizes these patterns in a limited-commitment model of marital investment, and empirically tests the proposed mechanism using a dynamic difference-in-differences strategy.

Drawing on multiple data sources, we first document marked trend breaks in marital decisions that coincide closely with the introduction of divorce. However, the direction of these breaks indicates an increased - rather than a diminished - appeal of marriage and marital specialization

resulting from the reform. Civil registry data on all marriages and births from to 1990-2018 reveal that, after declining quickly in the decades prior to legalization, the annual number of marriages stabilized in the year immediately following and remained stable for at least 15 years.

Following family economics theory, which links the gains from marriage to higher marital investments and marital specialization, we then examine trends in fertility and female labor supply. Similar to marriage rates, the fertility rates of married women declined quickly in the 1990s and the early 2000s, and started increasing after divorce was introduced. In contrast, fertility rates for unmarried women remained stable. This fertility rebound is also concentrated among married college-educated women. Consistently, their labor supply decreased after the law relative to women with lower schooling. Finally, we document that the share of marriages contracted under community property, which declined from 85% in 1990 to 55% in 2004, stopped declining after the law took effect.

Based on these trends, we hypothesize that, in community property marriages, the equal division of assets upon divorce insures homemakers against the economic consequences of marital breakdown, making women more likely to have children and reduce their labor supply. Before the law, community property assets could only be divided under very restrictive circumstances and remained under the husband's control otherwise.

The bulk of the empirical literature on divorce focuses on the conditions and access to divorce, studying whether more liberal access to divorce increases its likelihood and how the change in divorce risk, in turn, impacts marital decisions. We follow a smaller literature that is concerned with how family law divides marital property between ex-spouses. For example, the role of marital assets as an insurance mechanism has been studied within models of limited commitment and imperfect intra-household risk sharing in the context of the United States by Voena (2015) and by Lafortune and Low (2023). We adapt Lafortune and Low (2023)'s two-period model to the Chilean context to understand the effect of divorce legalization on marital decisions. In the model, equal asset division allows the husband to commit ex ante to compensating the wife after a divorce, which leads to increased specialization and higher gains from marriage.

To causally test this mechanism, we estimate the law's impact on marital specialization, proxied by married women's employment rates, using young women married before the law under community property as a treatment group, relative to a control group of women married under asset

separation. While both groups are exposed to the change in the availability of marital dissolution, which opens up remarriage opportunities, the division of assets upon divorce only affects women under the community of assets, making this comparison ideal for isolating the mechanism we are testing. We measure strong reductions in labor supply in the treatment group starting in the year of the reform, but only among college-educated women, consistent with the macro trends described earlier.

The model implies that the law should not impact women with a higher earning potential more strongly *ceteris paribus*. However, it also predicts a higher impact for households with higher asset levels, which are empirically correlated with income. This suggests that educational heterogeneity in macro patterns and reform impacts operate through differences in accumulated household assets, which are empirically correlated with female education. Accordingly, the treatment heterogeneity analysis reveals that the reform's impact is concentrated on households that had accumulated assets at the time of the law, as predicted by the model.

Overall, our results indicate that the legalization of divorce had a positive impact on marriage and fertility rates. In addition to the mechanism we studied, several aspects of the law, such as alimony provision or the ability to remarry, may have contributed to these patterns. Because both our treatment and control groups were exposed to these factors, our empirical strategy is effective in isolating the role of marital property division, and we measure large labor supply impacts. For the same reason, however, our exercise is silent on the quantitative contribution of these other factors to the macro trend breaks. However, the fact that marriage rebounds are concentrated in community property marriages suggests that marital asset division is a quantitatively important piece of the story.

This study contributes most directly to the small literature on divorce legalization by documenting a set of counterintuitive patterns and identifying a mechanism that was absent from existing work. The literature studying this fundamental policy change is limited because most countries legalized divorce long before micro household surveys were collected. In addition, divorce legalization is usually a country-wide change, which limits exploitable regional variation. Two studies specifically considered the short-term effects of Chile's 2004 Civil Marriage Act. Gallegos and Ondrich (2017) estimate a hazard model that identifies a relative increase in fertility in the five years following the law. We extend these results using multiple data sources spanning 30 years

and connecting it to large changes in marital outcomes at the population level.

The remainder of this paper is organized as follows: Section 2 discusses the implementation of divorce legalization in Chile. Section 3 describes the data sources used in this study. Section 4 analyzes trends in marital decisions and investments between 1990 and 2020 using civil registry data. Section 5 presents the hypothesized mechanism in a framework adapted from Lafortune and Low (2023) to the context of Chile. Section 6 estimates a dynamic difference-in-differences model to determine the causal mechanism. Section 7 concludes.

2 Background: Divorce Legalization in Chile

2.1 "Divorce" in Chile before 2004

Prior to 2004, Chile's marriage regime was governed by the Civil Marriage Act of 1884, which defined "divorce" in the following way: "*El divorcio no disuelve el matrimonio, sino que suspende la vida común de los cónyuges*" (Divorce does not dissolve marriage but rather discontinues the spouses' life together). This procedure was known as "divorcio no vincular" (in contrast to "divorcio vincular" introduced in 2004) and was akin to a legal separation: the marital bond remained intact, and remarriage was not permitted. Under "divorcio no vincular," marital obligations, including spousal support, subsisted. Such support was determined by the courts based on the needs of the recipient and the financial capacity of the obligated spouse, often considering fault in the separation. Regardless of the spouses' relationship, the law also recognizes child support obligations (Turner, 2004).

The only legal ways to dissolve a marriage were death or annulment. One of the most common grounds for annulment was the lack of legal formalities, such as errors in the identity of one spouse. Annulment required mutual consent if the marriage was not void from the outset, the sworn testimony of three witnesses to support the invalidating claim, and the involvement of a lawyer in the annulment process. It was prohibitively expensive and therefore concentrated in wealthier areas; anecdotal evidence suggests it could cost the equivalent of five months of minimum wage (Cox, 2011). Figure 1 shows the total rate of marital dissolution (comprising annulments and divorces after 2004) in Chile over time. The vertical dashed line in 2004 marks the enactment of the divorce law. It shows that annulment rates prior to the law were five or six times lower than the rate at which divorces eventually stabilized after the law.

2.2 The 2004 Civil Marriage Act

Law No. Law No.19.947, known as the New Civil Marriage Act and informally referred to as the Divorce Law due to its most significant reform, was enacted in 2004 and introduced legal divorce in Chile, allowing married couples to dissolve their legal bond and remarry. It permits the termination of a marital bond on three grounds: mutual agreement due to separation, unilateral petition due to separation, and fault-based divorce. Although divorce is not immediate under any of these options, the process usually concludes within a reasonable timeframe. In the first option, the couple must have been separated for at least one year, and the legal process typically lasts between 10 days and four months. In the second, the required separation period is three years, with proceedings usually taking four months to one year. In the case of fault-based divorce, there is no required separation period, but the judicial process tends to be longer and more complex.[3] To ease the procedural burdens and legal complexities of divorce, the public entity *Corporación de Asistencia Judicial* (Legal Aid Corporation) offers free legal guidance and, if needed, legal representation for those unable to afford a lawyer.

Another key change introduced by the law is economic compensation (alimony), a legal entitlement for a spouse who, due to childcare or household responsibilities, was unable to engage in paid work or did so to a lesser extent than they could and desired, resulting in economic disadvantage at the time of divorce or annulment. However, the law makes no distinction in terms of economic compensation between the two marital regimes. As Corral (2007) observes, compensation is based solely on the economic detriment suffered upon divorce, regardless of the divorce regime.

2.3 Timing of Legal Divorce Adoption

Chile's move from divorcio no vincular to legal dissolution of marriage (divorcio vincular) unfolded over nearly a decade, but the information environment relevant to the impacts we are investigating shifted sharply in 2003–2004. The divorce bill was first introduced in the Chamber of Deputies in November 1995, and although it advanced intermittently, years of legislative drift likely kept the perceived probability of near-term reform low for most households.[4]

The Senate finally approved the bill in principle on 13 August 2003, a highly reported milestone that opened a formal window for amendments and signaled that passage was no longer merely

[3] According to Serrano & Cía., a local family law firm.

[4] Senado de Chile, "Leyes con historia: El nuevo matrimonio civil".

4.1 Trends in Marriage Decisions

We first examine population-level data obtained from Chile's civil registry on all marriages celebrated in Chile between 1990 and 2018. The raw number of marriages in Chile exhibited a steady decline prior to 2004, as depicted in Figure 3a. This decline dates back to the late 1980s (Cox A. (2011)) and is consistent with the "retreat from marriage" observed in OECD countries and in Latin America since the 1970s, characterized by lower marriage rates and a substitution towards long-term cohabitation (Lundberg and Pollak, 2015; Wiegand-Cruz, 2025). Explaining this global trend is beyond the scope of this paper but we discuss in the conclusion the implications of our results for this broader debate.[12] After 2004, when divorce was introduced, the decline in marriages halted, and the annual number of marriages began to increase slowly over the next 15 years. Here, we only consider data for first-order marriages (in which neither spouse was married before). Therefore, the rebound in the flow of marriages does not correspond to a higher "churn" of marital relationships but rather an increased number of individuals choosing marriage over singlehood or cohabitation.

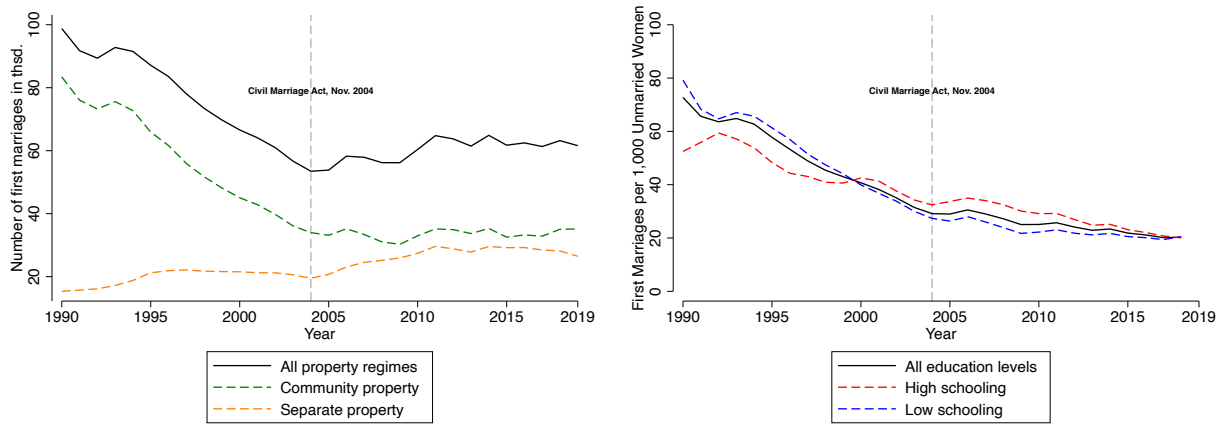
Figure 3a shows that the decline and subsequent rebound in the number of marriages can be attributed more specifically to community property (CP) than to separate property (SP) marriages. [13] The number of SP marriages grew steadily throughout the period, with perhaps a slight acceleration in 2004. Overall, while over 80% of new marriages were signed under CP in 1990, around half of newly married couples opted for this regime by 2004. This proportion remained stable thereafter. The choice of marital regime has received little attention in the literature, but a growing choice of separation property regimes has been documented in Italy and France (Frémeaux and Leturcq, 2013, 2018; Bayot and Voena, 2015), and is linked to reductions in the gender wage gap and growing female labor force participation. The patterns around Chile's 2004 Civil Marriage Act suggest

[12] Multiple mechanisms have been invoked to explain the decline in marriage rates, including changes in social norms around gender roles and marriage in connection to improved fertility control and higher female labor market participation (Cherlin, 1992; Akerlof et al., 1996; Goldin and Katz, 2002), changes in sex ratios (Angrist, 2002; Knowles and Vandenbroucke, 2019; Bronson and Mazzocco, 2024), a lack of "marriageable men" due to worsening economic opportunities, higher incarceration rates, or demographic shifts (Wilson, 2022; Ellwood and Crane, 1990; Charles and Luoh, 2010), and changes to the nature of the gains from marriage due to home production technological change (Greenwood et al., 2016; Lundberg and Pollak, 2015). Other papers point more specifically to the substitution between marriage and non-marital cohabitation due to the lower social cost of divorce or an increase in legal protections in cohabiting unions (Matouschek and Rasul, 2008).

[13] In the data, we observe a third asset regime of joint profit ("Participación en los gananciales") starting in 1995 - a hybrid between community assets and separation of assets. This asset regime is quantitatively negligible, and the fraction of marriages under this regime does not change meaningfully over time.

that the law significantly improved the appeal of the CP marriage regime but had a much smaller effect on the SP marriage regime. Therefore, we hypothesize in Section 5 that the counterintuitive impacts of the law are related to dispositions around the liquidation of the community property following a CP marriage dissolution.

Figure 3: Marriages in Chile 1990-2019, by Property Regime and Education



(a) Raw Number of Marriages by Property Regime

(b) Marriage Rate by Education Level

Notes: The left-hand panel figure shows the raw count of marriages by property regime. The data source is the Civil Registry. The right-hand panel shows the marriage rate excluding remarriages per 1,000 unmarried women. The marriage rate is defined as the number of new marriages divided by the number of unmarried women. The data source is the Civil Registry. The number of unmarried women was estimated from the CASEN in the survey years. For the intermittent years when no survey was conducted, we used interpolation methods. The sample is restricted to wives aged–20-39.

4.2 Trends in Marital Specialization

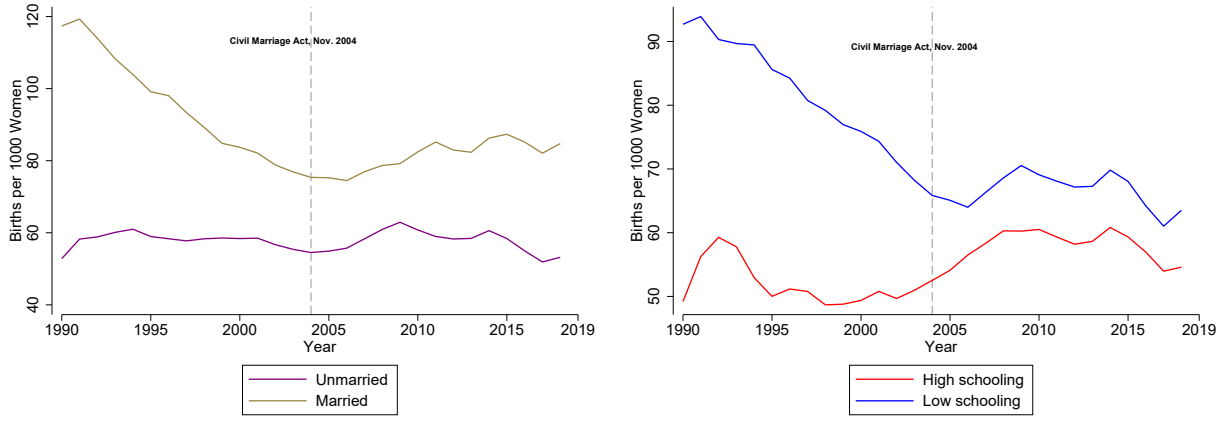
The ability of spouses to specialize in home production versus market work constitutes one of the main sources of marital surplus identified in the family economics literature (Weiss, 1997). Other sources, such as companionship or economies of scale in consumption, can also be achieved through cohabitation and could not have easily been positively altered by the introduction of divorce. As home production technologies have reduced the time required to produce marital commodities (such as meals, clean dwellings, and clothes), the value of marriage increasingly resides in the ability to invest in children's education (Greenwood et al., 2016; Lundberg and Pollak, 2015). Therefore, an increase in marriage rates should be associated with higher marital specialization and fertility.

Consistent with this argument, the pre-2004 fertility trends mirrored the declining marriage

trends. Figure 4a depicts the birth rates of married and unmarried mothers. Prior to 2004, birth rates markedly declined for married mothers and remained stable for unmarried mothers. After 2005, the declining trend in marital fertility stopped, and we observe a rebound in birth rates. These results imply that fertility did not rebound mechanically as a result of higher marriage rates (composition effect). Instead, fertility increased upon marriage.

In Figure 4b, we plot birth rates over time for women with at most a high school diploma versus women with at least some college education. For both groups, the introduction of divorce coincides with an increase in fertility, but the pre-reform fertility decline is concentrated among women with low schooling, whereas the rebound is more marked for women with more schooling.

Figure 4: Fertility by Mother's Marital Status and Education



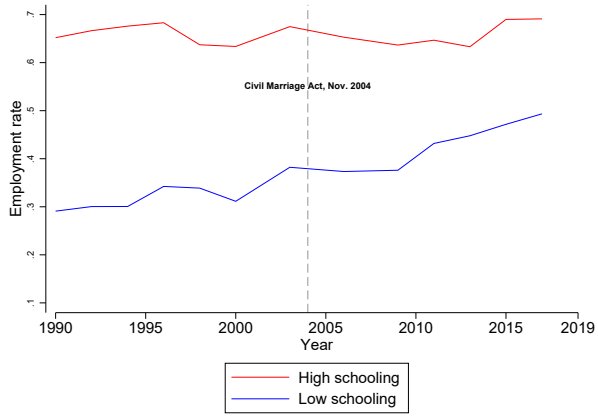
(a) Fertility Rate by Marital Status

(b) Fertility Rate by Education

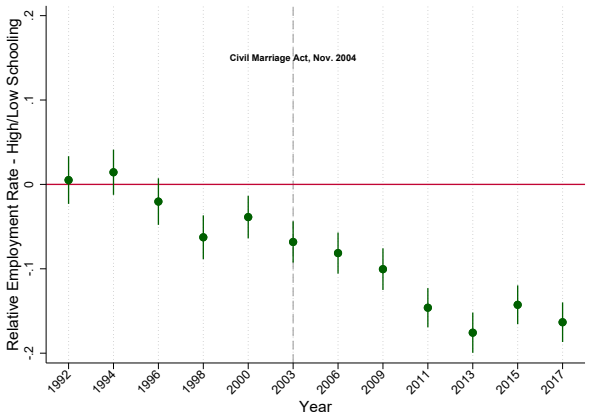
Notes: The left-hand panel figure shows the fertility rate by the mother's marital status. The right-hand panel shows the fertility rate by the mother's education level. The fertility rate is defined as the number of new births divided by the number of women. The data source is the Civil Registry. The number of women was estimated from CASEN in the survey years. For the intermittent years when no survey was conducted we use interpolation methods. The sample is restricted to women aged between 16-44.

Finally, we examine whether women's labor supply in Figure 5. Since labor supply and fertility are two correlated manifestations of marital specialization, we should expect consistent patterns between the two series. The left panels show the raw series for women's employment rates and hours worked (conditional on employment), by schooling. Trend breaks around 2004 are not as manifest as was the case for fertility rates, perhaps because labor supply is determined by other factors (e.g. macroeconomic conditions, structural transformation, technological changes, social

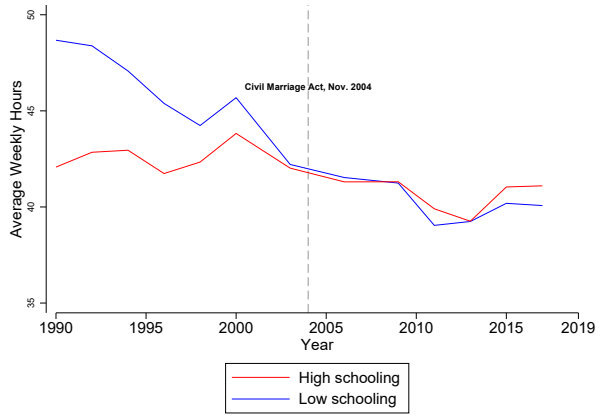
Figure 5: Female Labor Supply by Education



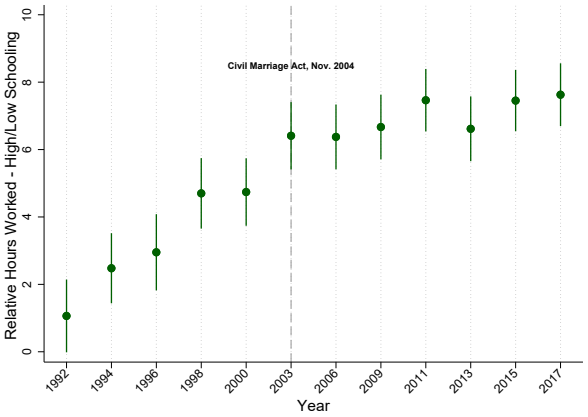
(a) Female Employment Rate by Education



(b) Effect of Schooling on Employment rates (base = 1990)



(c) Women's Average Weekly Hours Worked by Education



(d) Effect of Schooling on Weekly Hours Worked (base = 1990)

Notes: Panel (a) shows the raw female employment rate for women with high ($\geq$ 12 years) and low ($<$ 12 years) schooling. Panel (b) reports the interaction coefficients from equation (1) for employment, measuring the differential change for highly educated women relative to less educated women, with 1990 as the omitted year. Panel (c) shows the raw average weekly hours worked by education group, and panel (d) reports the corresponding coefficients for hours. The sample is restricted to women aged 25–44. The data source is the CASEN. .

remarry.

We use data from the EPS survey, a longitudinal household survey collected since 2004 with retrospective relationship histories and information on marital property regimes.[18] Our main sample includes women who were 20–29 years old and married in 2000; therefore, the reform impacted them at an age at which they were making fertility and career investment decisions. The panel dimension of the survey is crucial for identification, as explained below.

The survey is linked to the respondents' lifetime administrative pension contribution records, from which high-quality monthly formal employment status and earnings information can be observed. These two variables are our measures of marital specialization for this empirical exercise (higher female employment rates and earnings proxy lower marital specialization). A caveat to this choice of outcomes is that we cannot ascertain whether changes in formal employment reflect substitution with home production or child-rearing versus simple changes in leisure time or even formal versus informal employment substitution. Ideally, we would like to run our analysis on measures of time spent in home production or fertility, but the former is not recorded in the survey. While births are observed in the survey, they are rare events, which severely limits the power of our estimator given the available sample size.[19] Therefore, our interpretation of employment reductions as evidence of marital specialization relies on the fact that the timing and schooling heterogeneity of the impacts we estimate on employment coincide closely with the macro trends in fertility documented at the population level immediately after the reform passed (see Section 4).

We estimate treatment effects using Callaway and Sant'Anna's (2021) DiD framework in the single adoption date case (i.e., all treated units become treated in November 2004, denoted as t_0). We define treatment status as $D_{it} = 1\{i \in \mathcal{CP}\} \cdot 1\{t \geq t_0\}$, where $\mathcal{CP}$ is the set of women married under the community property regime, and estimate $\widehat{ATT}(t)$ for each post-period $t \geq t_0$ until December 2015, using women in separate asset marriages as the comparison group. The standard errors are two-way clustered at the individual and period levels to account for serial correlation from repeated observations. We implemented the estimator in Stata using the csdid package and confirmed the results by implementing OLS (after partialing out fixed effects using the reghdfe package) on the following model:

[18] The survey and sample restrictions are described in more detail in Section 3.

[19] Likewise, our exercise is underpowered to detect impacts of the reform on other outcomes of interest such as separation or divorce decisions.

$$Y_{it} = \alpha_i + \gamma_t + \sum_{k=-3}^{K=12} \beta_k CP_i \cdot \mathbf{1}\{t = t_0 + k\} + \varepsilon_{it} \tag{4}$$

where Y_{it} is the outcome measured on woman i in period t; CP_i equals one if woman i is married under CP; α_i are individual fixed effects and γ_t are period fixed effects; t_0 denotes the last pre-reform period October 2004, so each β_k measures the impact of CP status in the k-th month prior or after the introduction of divorce in November 2004; and ε_{it} is an error term. The coefficients β_k therefore trace the reform's dynamic effects on women's labor supply, controlling for both time-invariant individual and regime-specific differences (ϕ_{CP}) and common time shocks (γ_t). In the main text, we present the results obtained using annualized data. We also run the model on quarterly employment data with similar results (see Appendix Figure A.1).

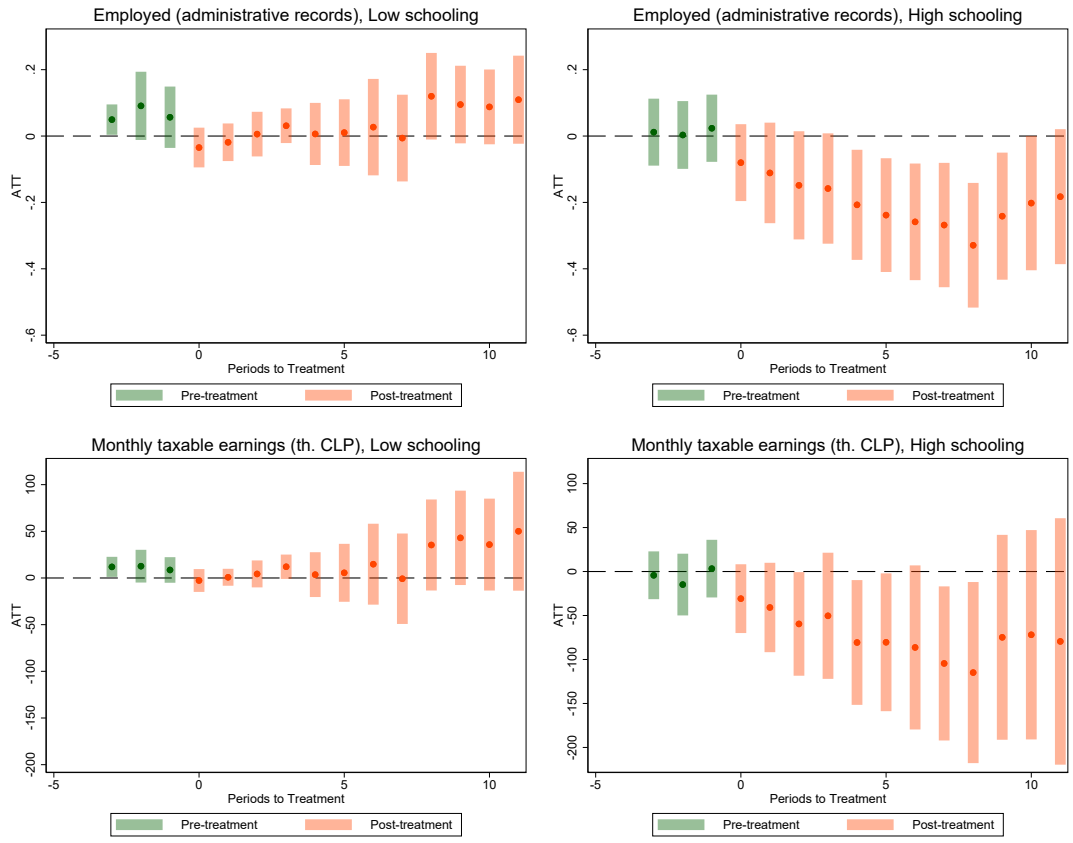
6.2 Threats to Identification

In addition to standard questions regarding the comparability of the treatment and control groups, which we discuss below, several identification threats need to be addressed in our context. First, the introduction of the divorce law affected marital decisions, including the choice of property regime. Consequently, the composition of the treatment and control groups may have changed in a way that could bias the before-and-after comparison within the groups. To address this issue, we leverage the panel dimension of the survey to extract a sample of women already married before the statutory reform under CP and SP rules. The sample comprises women who were already married in January 2000 and aged 20-29 years on that date.

Second, women in the treatment and control groups could exit marriage at different rates, modifying the relative composition of the two groups. Our main specification follows all women in the sample, regardless of whether they separated or divorced during the period. Therefore, the treatment effects capture both the direct effects of the reform on married women's specialization and the indirect effect mediated by the impacts of the reform on marital decisions. However, we also estimated an alternative specification only for women currently married in each period and found similar results, which suggests that indirect impacts play a minor role.

Differential attrition from the survey could also be an issue, particularly because marital transitions can be correlated with panel retention rates. To avoid this problem, we extracted our sample

Figure 7: Impact of Introducing Asset Division upon Divorce on Female Labor Supply



Notes: The figure reports event-study estimates from the Callaway–Sant'Anna DiD estimator (csdid) using individual–year data constructed from the EPS person-month panel and administrative records. Outcomes were collapsed to the individual–year level using annual means. The treatment timing was set to 2004. Non-employed women are assigned zero taxable earnings. The estimates are shown separately by education group (low vs. high schooling). The standard errors were clustered at the individual level. No panel fails the csdid pre-trends Wald test (a joint test of parallel trends that all pre-treatment event-time effects are zero), except for employment for the low-schooling group.

instrument that makes marital breakups more or less costly, its function as an insurance provision for homemakers has received much less attention, with some notable exceptions discussed in this paper.

While most countries allow for an equitable asset division upon marriage, understanding the broader lifetime penalties faced by women with children after a marriage breakup could help rationalize the current fertility decline observed worldwide, particularly as marital unions become less stable and rising female schooling and earnings potential make women less dependent on marriage for their economic wellbeing. For example, it is widely documented that alimony rights are often poorly enforced, despite the fact that parental responsibilities after divorce fall heavily on women. This suggests that the economic risk of divorce remains an important disincentive for fertility, even in countries that legalized divorce decades earlier.

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Rural Roads and Local Economic Development

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Abstract

Nearly one billion people worldwide live in rural areas without access to the paved road network. This paper measures the impacts of India's $40 billion national rural road construction program using regression discontinuity and data covering every individual and firm in rural India. The main effect of new feeder roads is to allow workers to obtain nonfarm work. However, there are no major changes in consumption, assets or agricultural outcomes. Nonfarm employment in the village expands only slightly, suggesting the new work is found outside of the village. Even with better market connections, remote areas may continue to lack economic opportunities.

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The Policy Research Working Paper Series disseminates the findings of work in progress to encourage the exchange of ideas about development issues. An objective of the series is to get the findings out quickly, even if the presentations are less than fully polished. The papers carry the names of the authors and should be cited accordingly. The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors. They do not necessarily represent the views of the International Bank for Reconstruction and Development/World Bank and its affiliated organizations, or those of the Executive Directors of the World Bank or the governments they represent.

Produced by the Research Support Team

I Introduction

Nearly one billion people worldwide live more than 2 km from a paved road, with one-third living in India (Roberts et al., 2006; World Bank Group, 2016). Fully half of India's 600,000 villages lacked a paved road in 2001. To remedy this, the Government of India launched the Pradhan Mantri Gram Sadak Yojana (Prime Minister's Village Road Program, or PMGSY) in 2000 with the goal of providing these villages with all-weather roads. Premised on the idea that "poor road connectivity is the biggest hurdle in faster rural development" (Narayanan, 2001) and promising benefits from poverty reduction to increased employment opportunities in villages (National Rural Roads Development Agency, 2005), the PMGSY has funded the construction of over 100,000 roads to 200,000 villages at a cost of almost $40 billion to date. Yet rural areas may have other disadvantages that may make it difficult to realize these gains; for example, they lack agglomeration economies and complementary inputs, such as human capital. Lowering transport costs may not be enough to transform economic activity and outcomes in rural areas.

Existing research is largely supportive of policymakers' claims: rural road construction is associated with increases in farm and non-farm economic growth as well as poverty reduction. But the causal impacts of rural roads have proven difficult to assess, mainly due to the the endogeneity of road placement. The high costs and potentially large benefits of infrastructure investments mean that the placement of new roads is typically correlated with both economic and political characteristics of locations (Blimpo et al., 2013; Brueckner, 2014; Burgess et al., 2015; Lehne et al., 2018). We overcome this challenge by taking advantage of an implementation rule that targeted roads to villages with population exceeding two discrete thresholds (500 and 1,000). This rule causes villages just above the population threshold to be 21 percentage points more likely to receive a road, allowing us to estimate the causal impact of rural roads using a fuzzy regression discontinuity design.

We construct a high spatial resolution dataset that combines administrative microdata on the universe of households and firms in rural India with remote sensing data and village aggregates describing amenities, infrastructure and demographic information. Because variation induced by program rules is across villages rather than across larger aggregates, and because of the possibility of heterogeneous effects by individual characteristics, village-identified microdata are essential for studying the impacts of roads. The limitation of this approach is that the administrative data are based on shorter questionnaires than traditional regional sample surveys.

In contrast with the dramatic economic benefits anticipated by policymakers, rural roads do not appear to transform village economies. Roads cause a substantial increase in the availability of transportation services, but we find no evidence for increases in agricultural production, assets or income. Farmers do not own more agricultural equipment, move out of subsistence crops or increase agricultural production. We can rule out a 10% increase in consumption with 95% confidence, with no significant or economically meaningful subgroup heterogeneity in terms of occupation, education or position in the consumption distribution.

We do find that rural roads lead to a large reallocation of workers out of agriculture. A new road causes a 10 percentage point decrease in the share of workers in agriculture and an equivalent increase in wage labor, measured on average four years after road completion. These impacts are most pronounced among the groups with the lowest costs and highest potential gains from participation in labor markets: households with small landholdings and working age men. We find suggestive evidence that reallocating workers are not the primary earners of the household, which may explain the limited effects on consumption.[1]

The growth in non-agricultural workers appears to be driven by work outside the village. We find small and insignificant increases in village non-farm employment (4 workers per

[1]Hicks et al. (2017) suggest an alternative explanation: rural-urban wage gaps are not as large as previously thought, at least for workers able to change occupation.

village), which can explain only 20% of the reallocation of workers out of agriculture. We also rule out changes in permanent migration, implying that the results we find are not the result of compositional changes to the village population.

In short, we find that the primary impact of new roads is to make it easier for workers to access outside labor markets. In the medium term at least, our research suggests that these external labor markets provide better opportunities than anything in their villages, even with high quality links to the road network. Roads alone appear to be insufficient to transform the economic structure of remote villages.

This paper contributes to a wide literature estimating the impacts of investments in transportation infrastructure. New highways and railroads have been shown to have substantial impacts on the allocation of economic activity, land use and migration.[2] But studies of major transportation corridors have limited external validity to the rural roads that we study, which connect poor, rural villages to regional markets. Existing research on rural roads in developing countries has used difference-in-differences and matching methods, largely finding positive impacts on both agricultural and non-agricultural earnings.[3] These studies are

[2]Trunk transportation infrastructure has been shown to raise the value of agricultural land (Donaldson and Hornbeck, 2016), increase agricultural trade and income (Donaldson, n.d.), reduce the risk of famine (Burgess and Donaldson, 2012), increase migration (Morten and Oliveira, 2017) and accelerate urban decentralization (Baum-Snow et al., 2017). Results on growth have proven somewhat mixed: there is evidence that reducing transportation costs can increase (Ghani et al., 2015; Storeygard, 2016), decrease (Faber, 2014) or leave unchanged (Banerjee et al., 2012) growth rates in local economic activity. Atkin et al. (2015) show that intra-country trade costs are very high in developing countries, with remote areas benefiting little from increased integration into world markets. For a recent survey of the economic impacts of transportation costs, see Redding and Turner (2015).

[3]Most closely related are papers that estimate the impact of rural road programs in Bangladesh (Khandker et al., 2009; Khandker and Koolwal, 2011; Ali, 2011), Ethiopia (Dercon et al., 2009), Indonesia (Gibson and Olivia, 2010), Papua New Guinea (Gibson and Rozelle, 2003) and Vietnam (Mu and van de Walle, 2011). Concurrent research on the PMGSY demonstrates that districts that built more roads experienced improved economic outcomes (Aggarwal, 2017), and that PMGSY roads lead to gains in agriculture (Shamdasani, 2016) and educational outcomes (Mukherjee, 2012; Adukia et al., 2017). Other papers also suggest that the lack of rural transport infrastructure may be a significant contributor to rural underdevelopment. Wantchekon et al. (2015) provide evidence that transport costs are a strong predictor of poverty across sub-Saharan Africa. Fafchamps and Shilpi (2005) offer cross-sectional evidence that villages closer to cities are more economically diversified, with residents more likely to work for wages. An older literature suggested that rural transport infrastructure was highly correlated with positive development outcomes (Binswanger et al., 1993; Fan and Hazell, 2001; Zhang and Fan, 2004), estimating high returns to such investments. Later

II Conceptual Framework

In this section, we sketch out a conceptual framework for understanding the impacts of new roads on village economies. Because we are interested in villages' productive structure, we explore impacts on occupational choice, agricultural production, and nonfarm firms. We focus on a set of channels that have received attention in existing research and in policymakers' justification for building rural roads.

The first order effect of a feeder road is to reduce transportation costs between a village and external markets, causing prices and wages to move toward prices outside the village. Given the sample of previously unconnected villages in India, this almost always implies higher wages, lower prices for imported goods and higher prices for exported goods.

We first consider farm production. A decline in the prices of imported inputs such as fertilizer and seeds can be expected to lead to greater input use and increased agricultural production. Changes in farmgate prices will cause crop choice to move in the direction of crops with the greatest price increases—those where the village has a comparative advantage. If agricultural production increases, it will also increase labor demand in agriculture, though these effects may be small or even reversed if production shifts to less labor intensive crops or if it becomes easier to import labor-substituting technology such as tractors.

The major offsetting effect is the increased access of village workers to external labor markets, which is likely to raise village wages. Higher labor costs will make farm work more expensive and may cause farms to reduce production and shift toward less labor intensive crops or technologies.

The impacts of roads on non-farm production in the village are analogous. Lower input prices and higher output prices will increase the production of non-farm goods, but these will be offset by higher wages. The relative changes in on-farm and off-farm production and labor demand will depend on the magnitude of the relative price changes between these

did not have paved roads, although in practice many projects under the scheme upgraded pre-existing roads. As the objective was to connect the greatest number of locations to the external road network at the lowest possible price, routes terminating in villages were prioritized over routes passing through villages and on to larger roads.

Importantly for this paper, national guidelines prioritized larger villages according to arbitrary thresholds based on the 2001 Population Census. The guidelines originally aimed to connect all villages with populations greater than 1,000 by 2003, all villages with population greater than 500 by 2007, and villages with population over 250 after that.[5] The thresholds were lower in desert and tribal areas, as well as hilly states and districts affected by left-wing extremism. These rules were to be applied on a state-by-state basis, meaning that states that had connected all larger villages could proceed to smaller localities. However, program guidelines also laid out other rules that states could use to determine allocation. Smaller villages could be connected if they lay in the least-cost path of connecting a prioritized village. Groups of villages within 500 m of each other could combine their populations. Members of Parliament and state legislative assemblies were also allowed to make suggestions that would be taken into consideration when approving construction projects. Finally, measures of local economic importance such as the presence of a weekly market could also influence allocation. Different states used different thresholds; for instance, states with few unconnected villages with over 1,000 people used the 500-person threshold immediately. Some states did not comply with the threshold guidelines at all. We identified complying states based on meetings with officials at the National Rural Roads Development Agency, which was the federal body overseeing the program.

Although funded and overseen by the federal Ministry of Rural Development, responsi-

[5]The unit of targeting in the PMGSY is the habitation, defined as a cluster of population whose location does not change over time. Revenue villages, which are used by the Economic and Population Censuses, are comprised of one or more habitations (National Rural Roads Development Agency, 2005). In this paper, we aggregate all data to the level of the revenue village.

To generate a measure of consumption, which is not directly surveyed by the SECC, we predict consumption in a district-level survey (IHDS-II, 2011-12) that contains the same asset, income and land data as the SECC. We then impute consumption for each individual in the sample following the small area estimation methodology of Elbers et al. (2003), allowing us to test not only for impacts of roads on mean consumption but also for distributional effects.[7] Appendix A contains additional details of this process.

Firm data comes from the Sixth Economic Census (2013). This covers every non-crop producing economic establishment in India, including public and informal establishments. It contains detailed information on location (which we match to the 2011 Population Census), employment, industry and a handful of other firm characteristics, but includes no variables on wages, inputs or outputs. We trim outliers to eliminate villages where the number of workers in village nonfarm firms is greater than the total number of workers resident in the village. Results are not substantively changed by this restriction.

Remote sensing data is used to measure outcomes otherwise unavailable at the village level. Nights lights provide a proxy for total village output. As no village-level agricultural output data exists in India, we use a satellite-based vegetative index (NDVI) for the primary (kharif) growing season (late May - October) to proxy for village-level agricultural production. To control for differences in non-crop vegetation, we measure the maximum growing season vegetation minus early cropping season vegetation.[8] We use village boundary polygons purchased from ML Infomap to map gridded remote sensing data to villages and to determine treatment spillover catchment areas.

The 2001 and 2011 Population Censuses provide village infrastructure, demographics, transportation services and the running variable for the regression discontinuity design (pop-

[7] Standard errors for all imputed consumption and poverty regressions are produced using the bootstrapping procedure outlined in Elbers et al. (2003).

[8] Table A1 shows that this measure is highly correlated with two other proxies for agricultural productivity and per capita consumption at the village level, as well as annual agricultural output at the district level. See Appendix A for additional details.

more aggregate levels is underpowered and faces greater identification concerns. We address these challenges by combining quasirandom variation from program rules with administrative census data georeferenced to the village level.

We obtain causal identification from the guidelines by which villages were prioritized to receive new roads. As previously described, new roads were targeted first to villages with population greater than 1,000, then those greater than 500, and finally greater than 250. While selection into road treatment may have been partly determined by political or economic factors, these factors do not change discontinuously at these population thresholds. As long as these rules were followed to any degree, the likelihood of treatment will discontinuously increase at these population thresholds, making it possible to estimate the effect of new roads using a fuzzy regression discontinuity design.

We pool villages according to the population thresholds that were applied in each state, so the running variable is village population minus the treatment threshold. Very few villages around the 250-person threshold received roads by 2012, so we limit the sample to villages with populations close to 500 and 1,000. Further, only certain states followed the population threshold prioritization rules as given by the national guidelines of the PMGSY. We worked closely with the National Rural Roads Development Agency to identify the state-specific thresholds that were followed and we define our sample accordingly. Our sample is comprised of villages from the following states, with the population thresholds used in parentheses: Chhattisgarh (500, 1,000), Gujarat (500), Madhya Pradesh (500, 1,000), Maharashtra (500), Orissa (500), and Rajasthan (500).[10]

Under the assumption of continuity of all other village characteristics other than road treatment at the treatment threshold, the fuzzy RD estimator calculates the local average treatment effect (LATE) of receiving a new road for a village with population equal to the

[10]These states are concentrated in north India. Southern states generally have far superior infrastructure and thus had few unconnected villages to prioritize. Other states such as Bihar had many unconnected villages but did not comply with program guidelines.

threshold. Following the recommendations of Imbens and Lemieux (2008) and Gelman and Imbens (2017), our primary specification uses local linear regression within a given bandwidth of the treatment threshold, and controls for the running variable (village population) on either side of the threshold. We use the following two stage instrumental variables specification:

$$\begin{aligned} Road_{v,j} = \gamma_0 + \gamma_1 1\{pop_{v,j} \geq T\} + \gamma_2(pop_{v,j} - T) + \\ \gamma_3(pop_{v,j} - T) * 1\{pop_{v,j} \geq T\} + \nu X_{v,j} + \mu_j + \upsilon_{v,j} \end{aligned} \tag{1}$$

$$\begin{aligned} Y_{v,j} = \beta_0 + \beta_1 Road_{v,j} + \beta_2(pop_{v,j} - T) + \\ \beta_3(pop_{v,j} - T) * 1\{pop_{v,j} \geq T\} + \zeta X_{v,j} + \eta_j + \epsilon_{v,j}. \end{aligned} \tag{2}$$

$Y_{v,j}$ is the outcome of interest in village v and group j, T is the population threshold, $pop_{v,j}$ is baseline village population, $X_{v,j}$ is a vector of village controls measured at baseline, and η_j and μ_j are district-threshold fixed effects. Village-level controls include indicators for presence of village amenities (primary school, medical center and electrification), the log of total agricultural land area, the share of agricultural land that is irrigated, distance in km from the closest census town, the share of workers in agriculture, the literacy rate, the share of inhabitants that belong to a scheduled caste, the share of households owning agricultural land, the share of households who are subsistence farmers, and the share of households earning over 250 INR cash per month (approximately 4 USD), all measured at baseline. District-threshold fixed effects are district fixed effects interacted with an indicator variable for whether the village is in the 1,000-person threshold group. $Road_{v,j}$ is an indicator that takes the value one if the village received a new road before the year in which Y is

Figure 4 shows the share of villages that received new roads before 2012 in each population band relative to the treatment threshold; there is a substantial discontinuous increase in the probability of treatment at the threshold. Table 2 presents first stage estimates using the main estimating equation at various bandwidths. Crossing the treatment threshold raises the probability of treatment by 21-22 percentage points; as suggested by the figure, the estimates are very robust to different bandwidth choices.

VI Results

VI.A Main results

To summarize the results, we begin by presenting treatment estimates on five indices of the major families of outcomes: (i) transportation services; (ii) sectoral allocation of labor; (iii) employment in nonfarm village firms; (iv) agricultural investment and yields; and (v) income, assets and consumption. We generate these indices to have a mean of 0 and a standard deviation of 1, following Anderson (2008); the variables that make up each index are described in the Data Appendix (Section A7). Table 3 presents the RD estimate of the impact of roads on each outcome. The first column shows a large positive effect on the availability of transportation services, and the second shows that roads cause a significant reallocation of labor out of agriculture. We find an almost significant positive effect ($p = 0.12$) on employment growth in village firms (Column 3), but small and insignificant positive effects on agricultural yields/investments and on the consumption index (Columns 4 and 5). These indices address concerns about multiple hypothesis testing within families of outcomes. To correct for cross-family multiple hypothesis testing, we follow the step-down procedure of Benjamini and Hochberg (1995), which allows us to reject the null hypothesis of zero effect on both transportation and agricultural labor share with a false discovery rate (adjusted p-value) of 0.05.

Figure 5 presents graphical representations of each column, showing the average of each

index as a function of distance from the treatment threshold. The plots show residuals from controls and fixed effects, along with linear estimations on each side of the threshold and 95% confidence intervals. The graphs corroborate the tables, showing significant treatment effects for transportation and labor exit from agriculture, but little clear impact on the firms, agricultural production and consumption indices. These results broadly summarize the findings of this paper: rural roads lead to increases in transportation services and reallocation of labor out of agriculture, but no major changes to village firms, agricultural production and consumption. The rest of this section examines the components of each of these indices to explain the impacts of roads in more detail, and presents results on heterogeneity.

Table 4 shows regression discontinuity estimates of the impact of a new road on an indicator variable for the regular availability at the village level of the five motorized transportation services that are recorded in the 2011 Population Census. A new road causes a statistically significant 12.8 percentage point increase in the availability of public bus services, more than doubling the control group mean of 11.9 percent. The impact on private buses is nearly as large but measured with less precision. Taxis and vans, which are more expensive forms of transportation, do not experience significant growth. Availability of auto-rickshaws, the least expensive private form of motorized transport, increases as well. Given that we are unable to observe transportation costs directly, we interpret these results as evidence that the new roads studied in this paper do meaningfully affect connections between treated villages and outside markets.[14]

Table 5 presents impacts of new roads on occupational choice, the one domain where roads appear to substantially change economic behavior. As 92% of workers in sample villages report their occupation to be either in agriculture or in manual labor, we focus our investigation on these categories. The first two columns show the impact of new roads on the

[14]This finding is not a given; Raballand et al. (2011) argue that in remote areas of Malawi, willingness to pay for transportation services may be so low that roads may not appreciably improve transportation options.

share of workers (aged 21-60) who work in agriculture, and who work as manual laborers. New roads cause a 10.1 percentage point reduction in workers in agriculture (representing a 21% decrease from the control group mean) and an 8.0 percentage point increase in workers in (non-agricultural) manual labor.[15] Columns 3 and 4 report estimates on the share of households deriving their primary source of income from cultivation and manual labor. In contrast to worker-level estimates, these regressions demonstrate that household income source does not change significantly, suggesting that many of the workers exiting agriculture are not the primary earners in the household. This result is also consistent with the finding that consumption and income do not change dramatically in response to new roads.

Theoretically, we should expect those who exit agriculture in favor of nonfarm labor market opportunities will be those for whom the losses of agricultural income are smallest and the labor market gains are largest. By using individual-level census data, we can examine the distribution of treatment effects across subgroups with different factor endowments. As land is the major input into agricultural production, land endowments may play a major role in determining which workers respond most to a rural road. We first examine the impact of road construction on the landholding distribution in Table A4. We find that a new road does not significantly change the share of households that are landless, own less than 2 acres, or have between 2 and 4 acres of agricultural land. However, we do find a 3.4 percentage point increase in the share of households with over four acres of land (significant at the 10 percent level). We are hesitant to over-interpret one marginally significant result out of four tests, but it is possible that there is some land consolidation following the construction of a road. Regardless, we do not find major changes in the landholding distribution and thus treat ex post observed landholdings as a baseline variable upon which to conduct heterogeneity

[15]The SECC does not report manual labor occupations in more detail. Table A3 breaks down the sectoral distribution of non-agricultural manual laborers using the 68th round of the National Sample Survey (2011-12). By far the most common category of manual labor in India is construction, making it a likely sector for many of these former agricultural workers.

interpret the observed sectoral reallocation of labor as the result of changes in occupational choice rather than compositional effects due to selective migration.

Table A10 addresses the possibility that the workforce has changed, which would make it difficult to interpret changes in the share of workers in agriculture or non-agricultural wage work. The table shows that roads do not affect the share of adults who are either not working or who are in occupations that we are unable to classify, suggesting that this potential bias is not important.

A different threat to our identification could come from any other policy that used the same thresholds as the PMGSY. In fact, one national government program did prioritize villages above population 1,000: the Total Sanitation Campaign (Spears, 2015), which attempted to reduce open defecation through toilet construction and advocacy. It is unlikely that this program is spuriously driving our results for two reasons. First, there is little theoretical reason to believe that investments in sanitation could drive large increases in transportation services or reallocation of labor away from agriculture. Second, in Table A11 we present regression discontinuity estimates of the impact of road prioritization on four measures of sanitation. We find no evidence that being above the population threshold is associated either with open defecation or any measure of access to toilets, suggesting that there is no discontinuity in the implementation of the program that might affect our results.

Finally, we consider the possibility that roads have spillover effects on nearby villages; if so, our estimates of direct effects could be biased either upwards or downwards relative to the total effects of new road provision. To do so, we examine outcomes in villages within a 5 km radius of villages in the main sample, using the standard regression discontinuity specification. Table A12 presents results of these regressions for the outcome family indices. We find no evidence of spillovers, and can reject equality with the main point estimates on

and highways. The construction of a paved rural road is unlikely to significantly change the one-time cost of permanent migration relative to the lifetime benefits, in contrast to the major changes induced by highway construction.

the transportation and agricultural occupation measures.

VII Conclusion

A large share of the world's poor are in rural areas without access to the paved road network. The resulting high transportation costs potentially inhibit gains from the division of labor, economies of scale and specialization.

In this paper we estimate the economic impacts of the Pradhan Mantri Gram Sadak Yojana, a large-scale program in India that seeks to provide universal access to paved "all-weather" roads in rural India. We exploit discontinuities in the probability of paved road construction at village population thresholds, finding that rural roads do not substantially change village economies. In agriculture, the largest sector in rural India, these roads affect neither input usage nor output. We do find that new paved roads lead to increased transportation services and a large reallocation of labor out of agriculture. But rather than causing nonfarm growth in rural areas, roads appear to facilitate the access of rural labor to external employment.

Roads are costly investments: the cost of connecting each village to the paved road network is approximately $150,000. In contrast to expectations that these roads would boost income and reduce poverty, we find insignificant and small effects on earnings and consumption. In our sample, the mean consumption per capita is approximately $267 per year and the average village has 696 inhabitants. We estimate an increase in consumption of 2.3%, which translates into $6.14 additional annual consumption per person, or only $4274 per year for the village as a whole. Even if we use the upper bound of the confidence interval on consumption, we find small effects relative to the cost of roads. This number appears even smaller when one considers that roads require costly ongoing maintenance. Worse yet, the villages in India still lacking paved roads are less populated and more remote than those in our sample, suggesting that impacts for future rural road investments are likely to be